

Wafer-Level IC Yield Analysis During Semiconductor Pilot Manufacturing Using JMP

Project Overview

- This project analyzes IC yield during the pilot manufacturing stage of a new semiconductor process.
- Two manufacturing methods are compared:
 - **Method A:** reference process
 - **Method B:** quicker and less expensive process
- The main goal is to determine which method gives better yield and whether Method B is suitable for further production development.
- JMP was used to analyze pass/fail test results, compare yield, and create wafer-level spatial maps.

Dataset Summary

- Total wafers analyzed: **50 wafers**
- Method A wafers: **25**
- Method B wafers: **25**
- ICs per wafer: **1,300**
- Total IC test records: **65,000**
- Each IC was electrically tested and classified as:
 - **Passed**
 - **Failed**

Project Objectives

The objectives of this project are to:

- Compare the overall yield of Method A and Method B.
- Calculate 95% confidence intervals for the yield of each method.
- Test whether the yield difference between Method A and Method B is statistically significant.
- Compare wafer-to-wafer yield variation within each method.
- Create wafer-level pass/fail maps using X-Y die coordinates.
- Identify spatial failure patterns across the wafer.
- Analyze center, middle, and edge wafer regions separately.
- Rank the best and worst wafers based on wafer-level yield.
- Provide an engineering recommendation.

Relevant JMP Platforms and Statistical Techniques

Distribution: Bar Chart, Summary Statistics, Confidence Interval for a proportion, Confidence Interval for difference in proportions, confidence interval for ratio of proportions.

Graph Builder: Mosaic Plot, Heat Map

The dataset includes six variables: Method, Wafer, Method-Wafer-IC, X Coordinate, Y Coordinate, and Test Result. Method identifies the process used to produce the wafer, while Wafer identifies the wafer number within each method. The X and Y coordinates describe the physical die location on the wafer surface. Test Result is the binary response variable used to calculate IC yield.

Overall Yield by Method A vs Method B

Which process method gives higher IC yield: **Method A** or **Method B**?

Yield means:

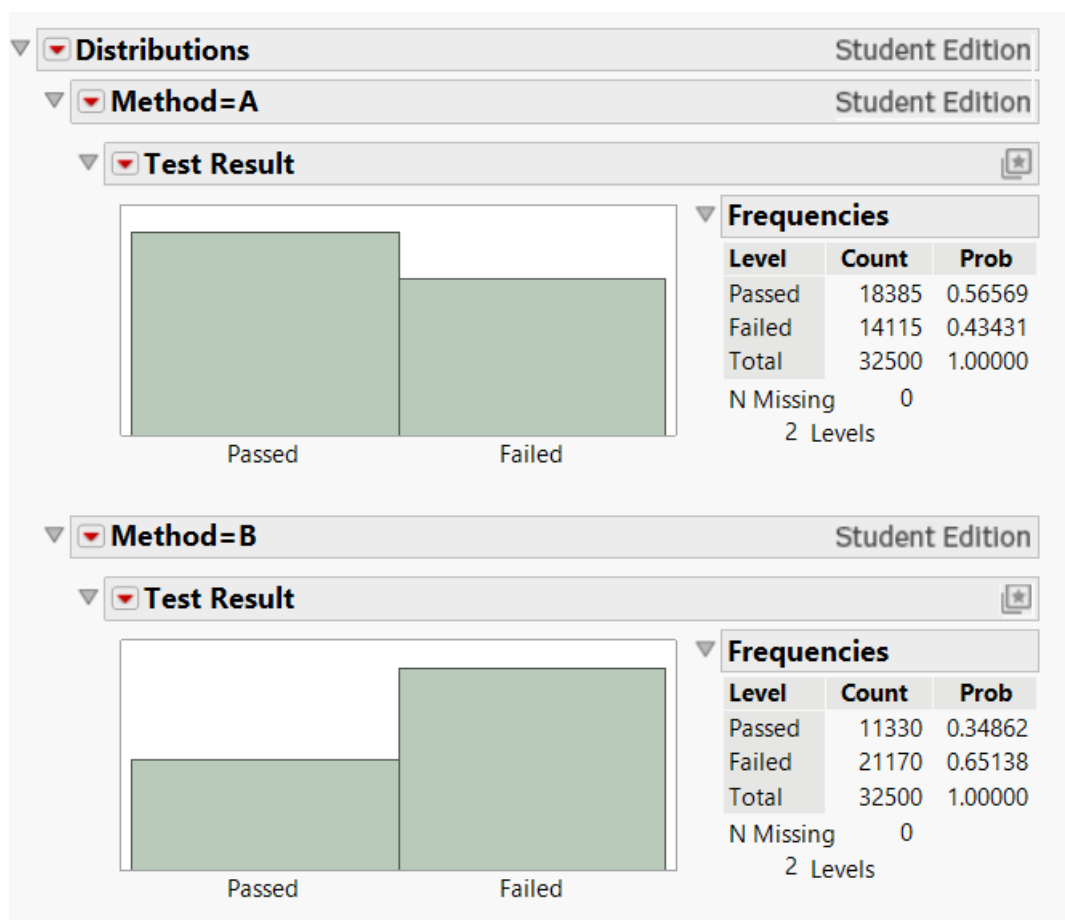
$$\text{Yield} = \frac{\text{Passed ICs}}{\text{Total ICs}} \times 100$$

For each method, total ICs should be:

$$25 \text{ wafers} \times 1300 = 32500$$

So Method A has 32,500 ICs and Method B has 32,500 ICs.

$$\text{Prob} = \frac{\text{Count for that result}}{\text{Total ICs for that method}}$$



Test Results of method A and B

The overall yield analysis shows that Method A produced a higher percentage of functional ICs compared with Method B. Method A had 18,385 passing ICs out of 32,500, corresponding to a yield of 56.57%. Method B had 11,330 passing ICs out of 32,500, corresponding to a yield of 34.86%. Therefore, Method A outperformed Method B by 21.71 percentage points. Although Method B is described as a quicker and less expensive process, the pilot manufacturing data shows a substantially lower yield, suggesting that Method B may require further process optimization before it can be considered for full-scale production.

So, the **yield of Method A** is: $0.56569 \times 100 = 56.57\%$, and the **yield of Method B** is: $0.34862 \times 100 = 34.86\%$

95% Confidence Interval for IC Yield by Manufacturing Method

This section estimates the likely range of true production yield for Method A and Method B using 95% confidence intervals. Since yield is based on the proportion of ICs that passed electrical testing, the confidence interval is calculated for the “Passed” category.

We do this because the yield values we got are from a **pilot sample**, not full-scale production. The confidence interval tells us the likely range of the **true long-term yield** if the method is used in manufacturing.

▼ Confidence Intervals					
Level	Count	Prob	Lower CI	Upper CI	1-Alpha
Passed	18385	0.56569	0.560296	0.571073	0.950
Failed	14115	0.43431	0.428927	0.439704	0.950
Total	32500				
Note: Computed using score confidence intervals.					
Student Edition					
▼ Confidence Intervals					
Level	Count	Prob	Lower CI	Upper CI	1-Alpha
Passed	11330	0.34862	0.343453	0.353814	0.950
Failed	21170	0.65138	0.646186	0.656547	0.950
Total	32500				
Note: Computed using score confidence intervals.					

Yield=Probability of Passed ICs

Method	Passed ICs	Total ICs	Yield	95% Confidence Interval
Method A	18,385	32,500	56.57%	56.03% to 57.11%
Method B	11,330	32,500	34.86%	34.35% to 35.38%

Method A showed a yield of 56.57%, with a 95% confidence interval from 56.03% to 57.11%. Method B showed a lower yield of 34.86%, with a 95% confidence interval from 34.35% to 35.38%. Since these intervals do not overlap and are widely separated, the results indicate a significant difference in yield between the two manufacturing methods. Method A outperformed Method B by approximately 21.71 percentage points. The confidence intervals are very narrow because each method was evaluated using 32,500 IC test results, giving a large sample size and therefore a small margin of error.

The 95% confidence intervals for both methods are very narrow because each method contains a large number of IC-level observations. Since 32,500 ICs were tested for each method, the estimated pass proportions have a small margin of error. This provides high statistical precision in estimating the yield for each method.

Method A has significantly higher yield than Method B.

The yield difference is: $56.57\% - 34.86\% = 21.71\%$

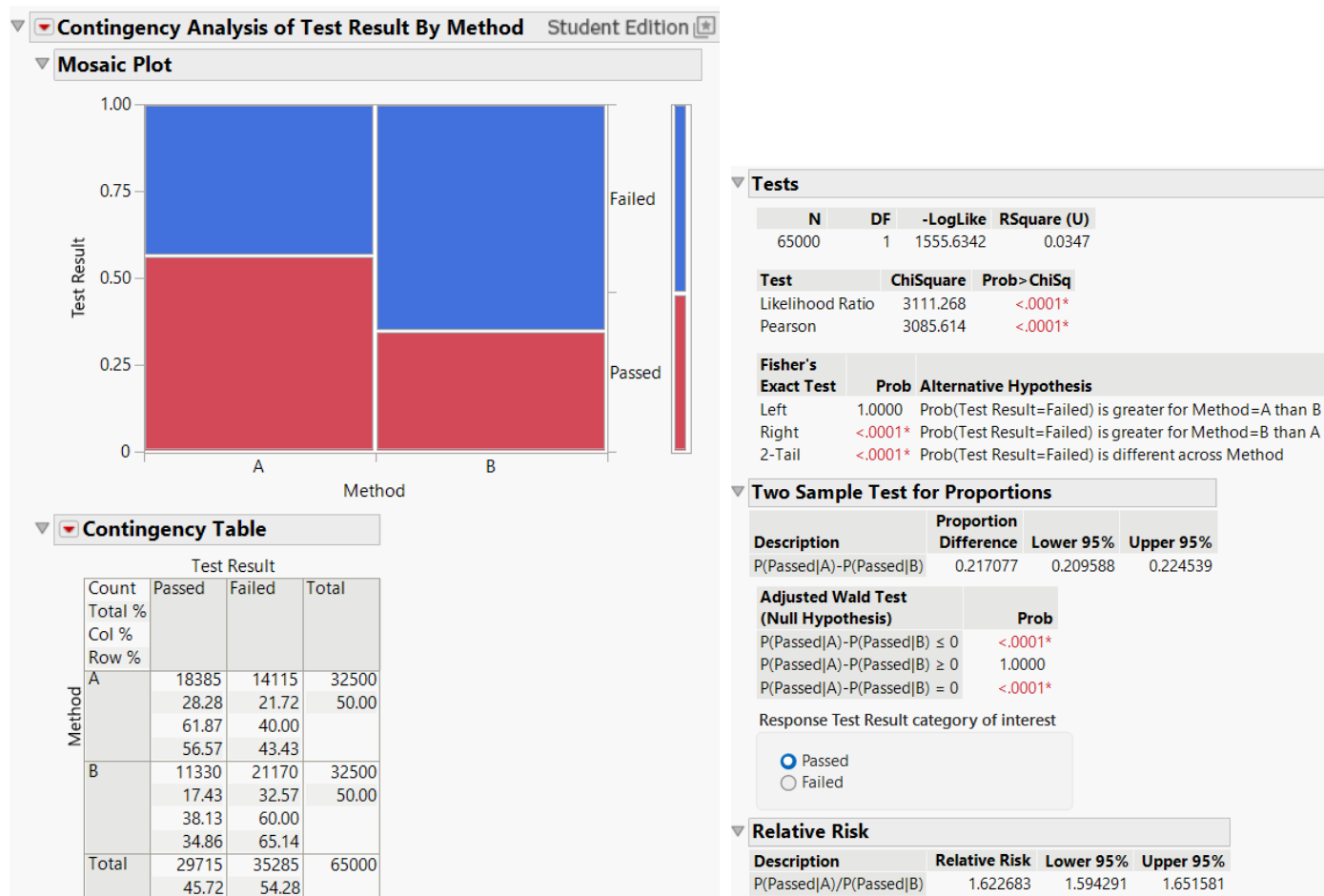
Statistical Comparison of Yield Difference Between Methods

So Method A looks better. But now we want to quantify the difference directly.

The yield difference is: $56.57\% - 34.86\% = 21.71\%$

This means Method A produced **21.71 percentage points more passing ICs** than Method B. But JMP will give the formal 95% confidence interval for that difference.

How much higher Method A yield is than Method B , The ratio of Method A yield to Method B yield ?



The mosaic plot is a visual comparison of **Passed** and **Failed** ICs for Method A and Method B. In plot: Method A has a larger **Passed** portion. Method B has a larger **Failed** portion. So visually, Method A clearly has better yield than Method B.

Method A produced: $18,385 - 11,330 = 7,055$, more passing ICs than Method B in the same sample size.

Method B has a significantly higher failure rate than Method A.

With 95% confidence, Method A's true yield is between **20.96 and 22.45 percentage points higher** than Method B's true yield.

Method A's yield is about **1.62 times** Method B's yield. The 95% confidence interval says: With 95% confidence, Method A's true yield is between **1.59 and 1.65 times** Method B's true yield

A two-sample test for proportions was used to directly compare the IC yield of Method A and Method B. The estimated difference in yield, calculated as $P(\text{Passed}|A) - P(\text{Passed}|B)$, was 0.2171, meaning Method A produced a yield that was 21.71 percentage points higher than Method B. The 95% confidence interval for this difference was 0.2096 to 0.2245, or 20.96 to 22.45 percentage points. Since this interval is entirely above zero and the p-value for equal proportions is less than 0.0001, the yield difference is statistically significant. The relative risk was 1.6227, with

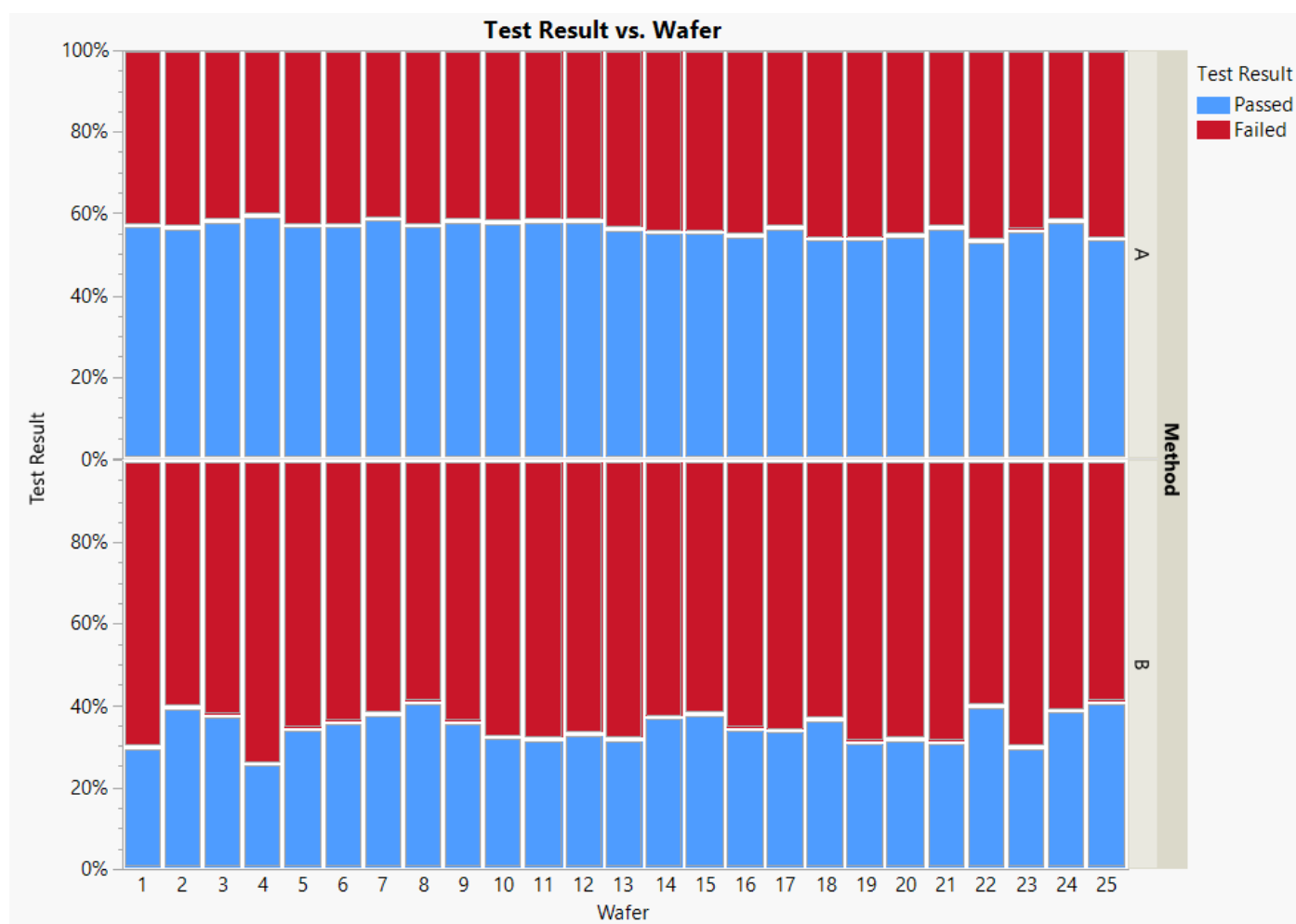
a 95% confidence interval from 1.5943 to 1.6516, indicating that Method A's yield was approximately 1.62 times Method B's yield. For engineering communication, the difference in yield is preferred because it directly describes the absolute yield loss and can be translated into additional passing ICs. In this pilot study, Method A produced 7,055 more passing ICs than Method B for the same number of tested ICs

Wafer-to-Wafer Yield Variation

We only compared **overall Method A vs Method B**. But that does not tell us whether the yield is stable across wafers. For example:

- Method A may have high average yield, but a few wafers may be very bad.
- Method B may have low average yield across almost every wafer.
- Some wafers may behave like outliers.

This step checks: Is the yield relatively consistent across the 25 wafers within each method?



Yes, the yield is **relatively consistent within Method A**, but **less consistent within Method B**. In this mosaic plot, each vertical bar represents one wafer. Since each wafer has **1,300 ICs**, the height of the blue section shows the **yield/pass percentage** for that wafer, and the red section shows the **failure percentage**.

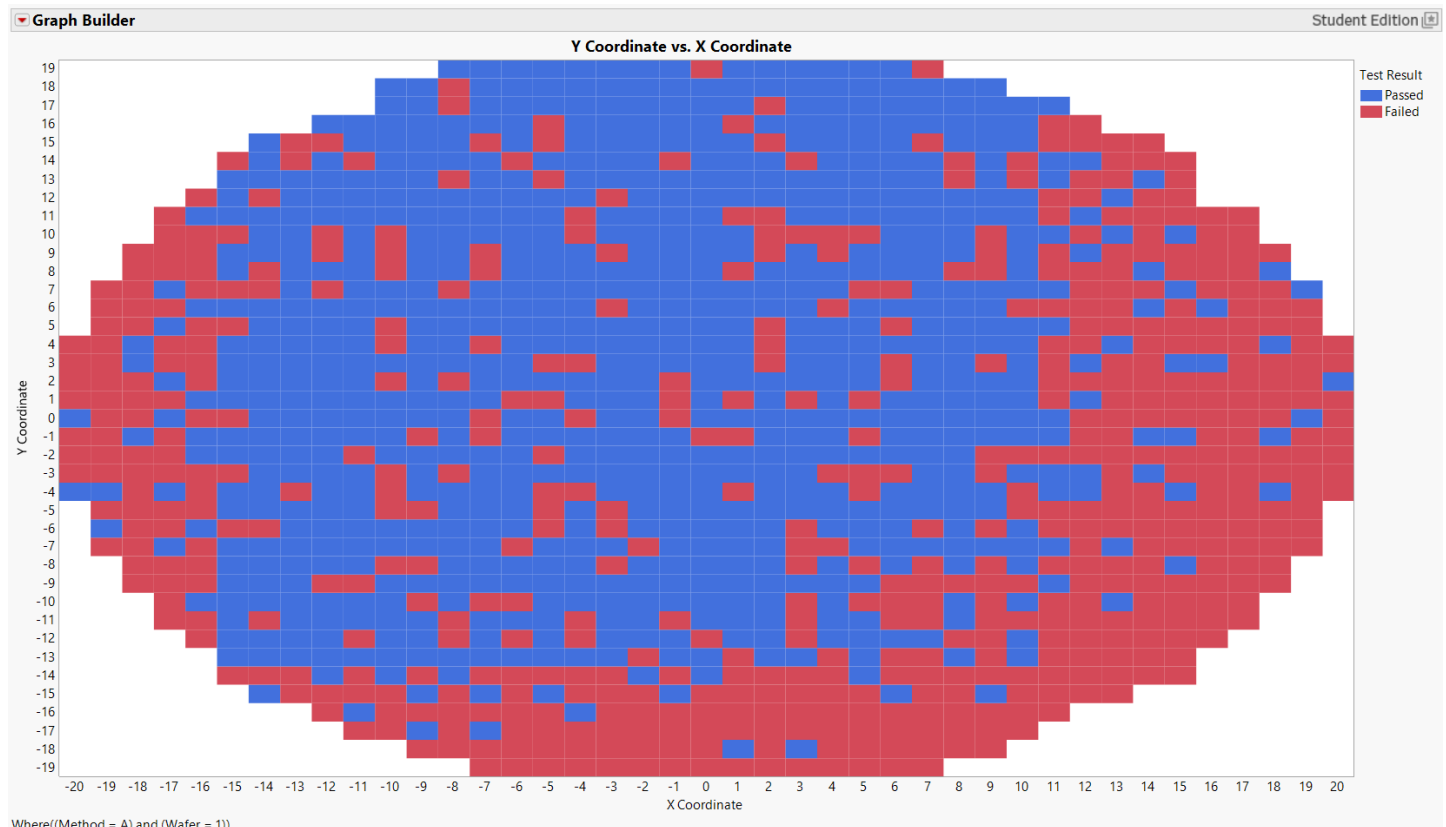
Method A shows a fairly stable wafer-to-wafer yield. Most wafers in Method A have a pass yield around: 55% to 59%. There are small differences between wafers, but no wafer looks extremely different from the others. Also, for every Method A wafer, the blue **Passed** portion is larger than the red **Failed** portion.

Method B has much lower yield overall and shows more wafer-to-wafer variation. Most wafers in Method B have pass yield around: 30% to 40%. Some wafers look especially poor, such as wafers around **4, 12, 19, 20, 21, and 23**,

where the blue Passed region is smaller. A few wafers, such as **8, 22, 24, and 25**, look slightly better, but still much lower than Method A. For every Method B wafer, the red **Failed** portion is larger than the blue **Passed** portion.

The wafer-to-wafer mosaic plot shows that Method A has relatively consistent yield across its 25 wafers, with most wafers showing a pass percentage around the mid-to-high 50% range. No major outlier wafer is observed for Method A. In contrast, Method B shows consistently lower yield across all wafers, with most wafer-level yields falling around the 30–40% range. Method B also appears to show larger wafer-to-wafer variation, with some wafers performing noticeably worse than others. Overall, the wafer-level analysis indicates that Method A is both higher yielding and more stable across wafers, while Method B shows a systematic yield loss across the wafer batch.

Wafer Map and Spatial Failure Pattern Analysis

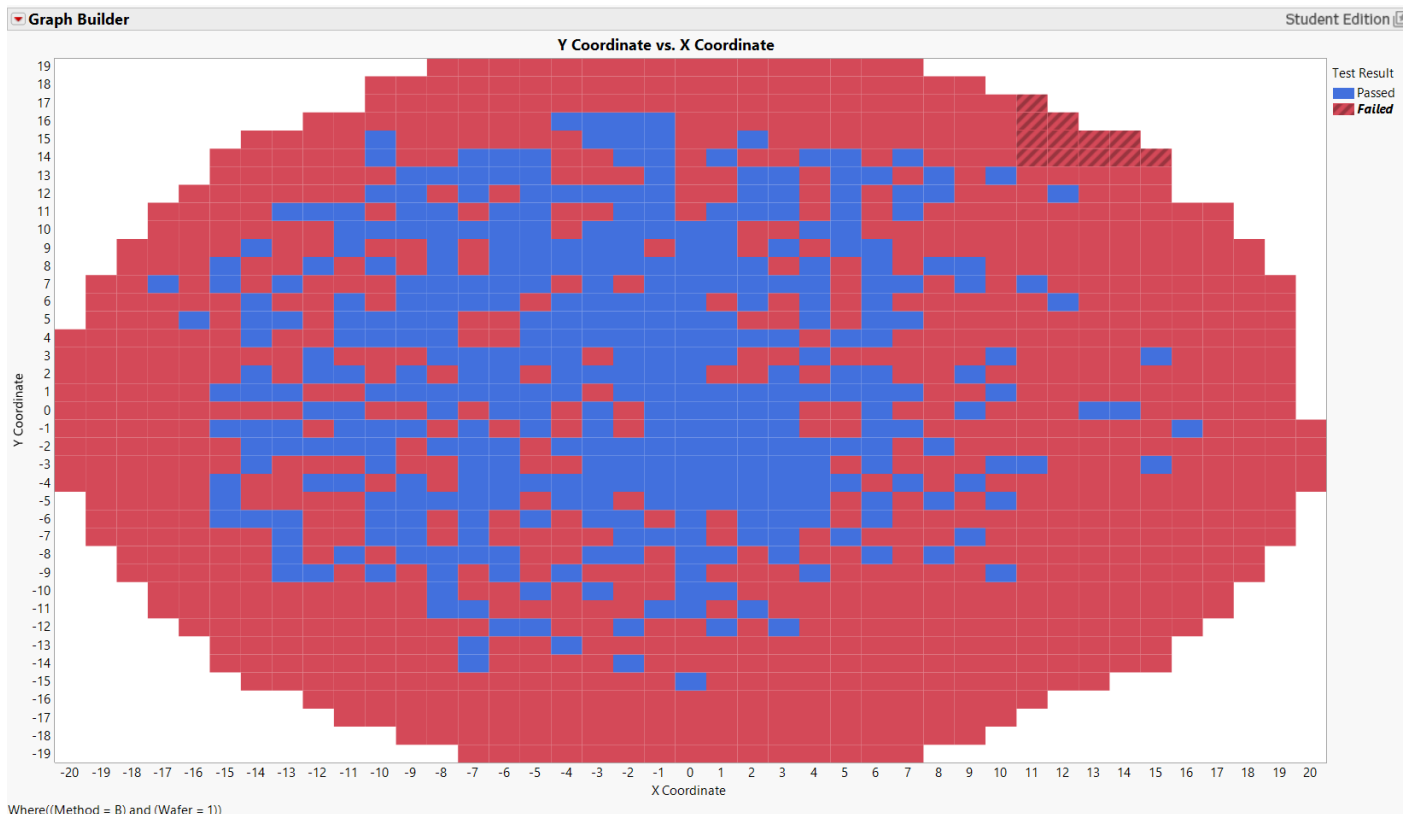


Heat Map for Method A wafer 1

Method A Wafer 1 has more blue regions than Method B Wafer 1, which matches our earlier result that Method A has higher yield. The main pattern is:

- The **central region** has many passed ICs.
- The **right edge** has a large failed region.
- The **bottom edge/lower region** also has many failures.
- Failures are not fully random; there is a clear spatial pattern.

So for Method A, the yield is better overall, but the wafer still shows **edge-related or regional failure patterns**, especially near the right side and lower edge. Failures increase near some wafer edges. This may indicate edge-related process nonuniformity, such as lithography, etch, deposition, thermal nonuniformity, or edge handling effects.



Heat Map for Method B Wafer 1

Method B Wafer 1 is much worse.

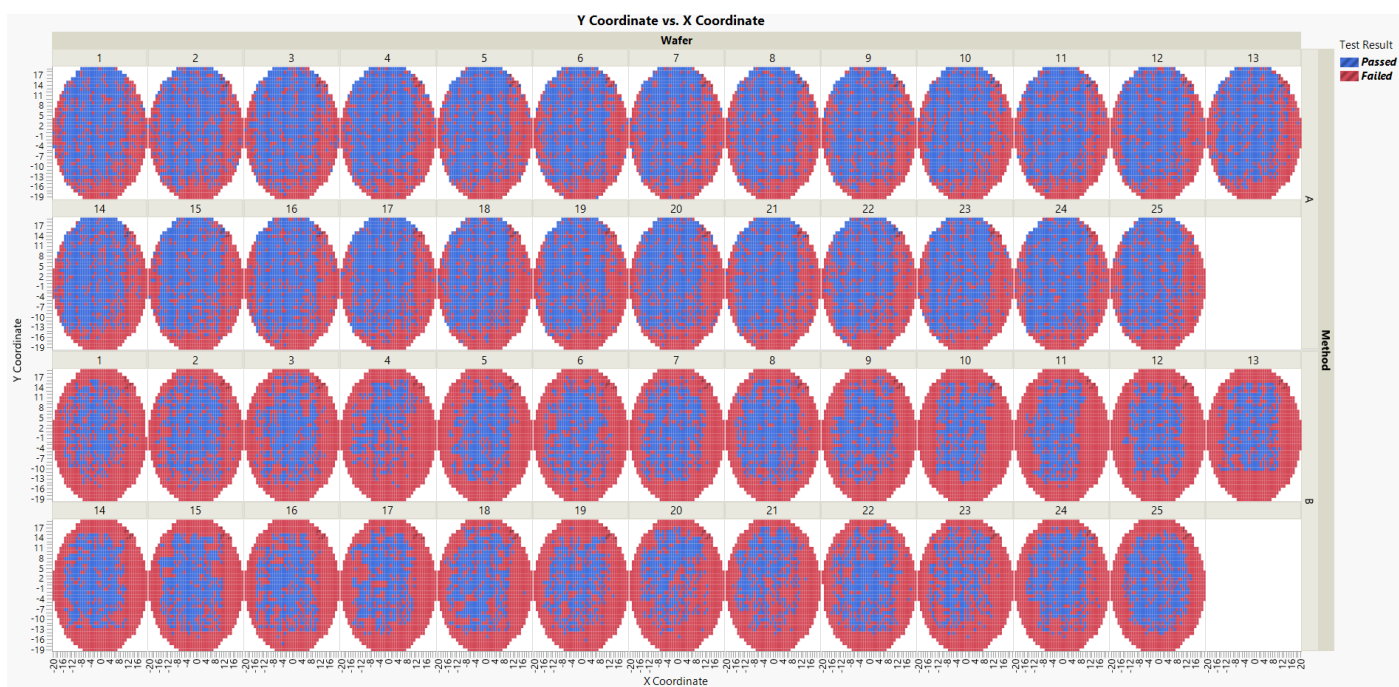
The main pattern is:

- Large portions of the wafer are red.
- The **outer edge region** is mostly failed.
- The passed ICs are mainly concentrated near the **middle/central region**.
- The red failure region is much larger than in Method A.

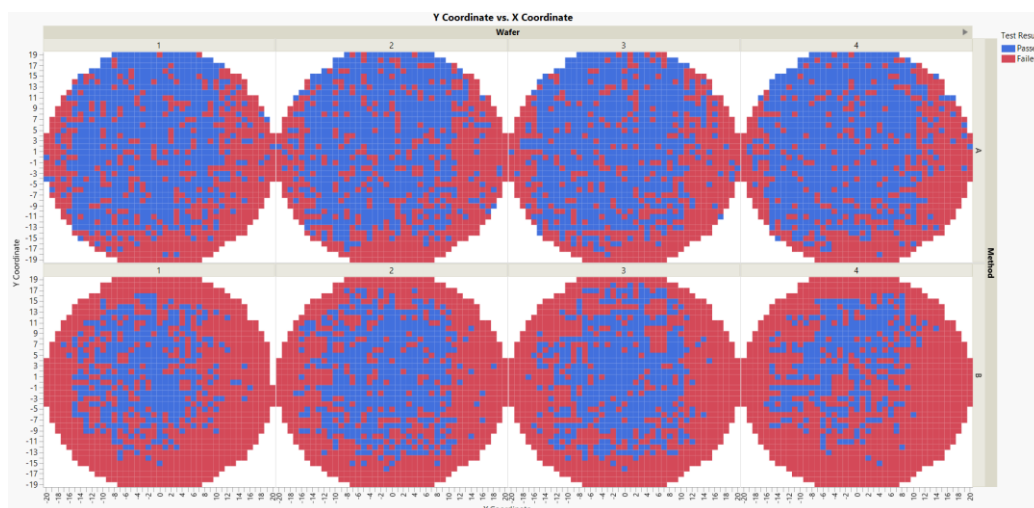
So Method B seems to have a stronger spatial yield problem.

Method B shows severe edge-related yield loss, with passing ICs mostly concentrated in the central area of the wafer. This suggests that Method B may have stronger process nonuniformity across the wafer compared with Method A.

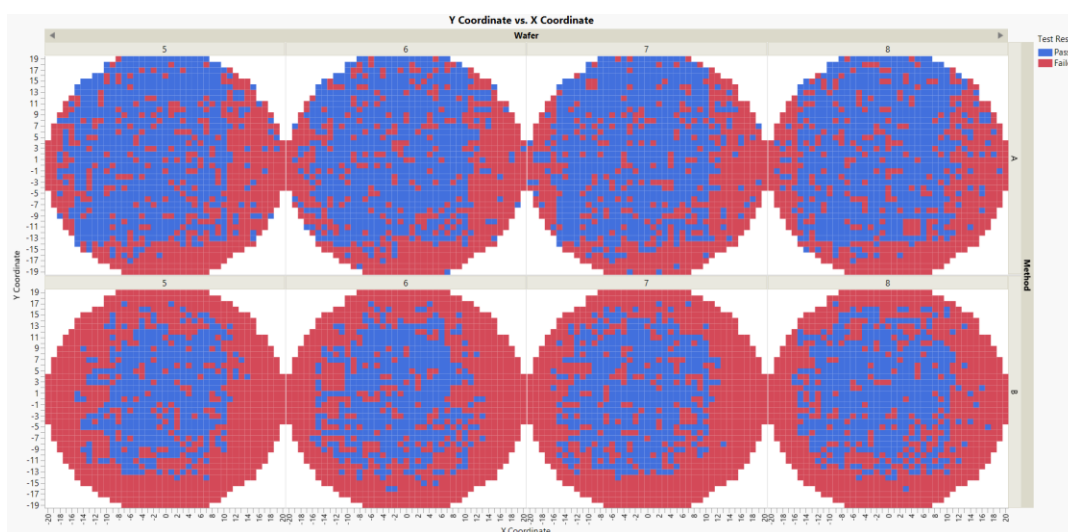
Representative wafer-level pass/fail heat maps were generated for Method A Wafer 1 and Method B Wafer 1. In these maps, each IC location is plotted using its X and Y coordinate, with blue indicating passed ICs and red indicating failed ICs. Method A Wafer 1 shows a larger passing region, especially near the central area of the wafer, although failures are still visible near the right and lower wafer edges. In contrast, Method B Wafer 1 shows a much larger failed region, with passing ICs mainly concentrated near the wafer center. This suggests that Method B may have stronger spatial nonuniformity and edge-related yield loss compared with Method A. However, the full all-wafer heat map is needed to determine whether this pattern is consistent across all wafers.



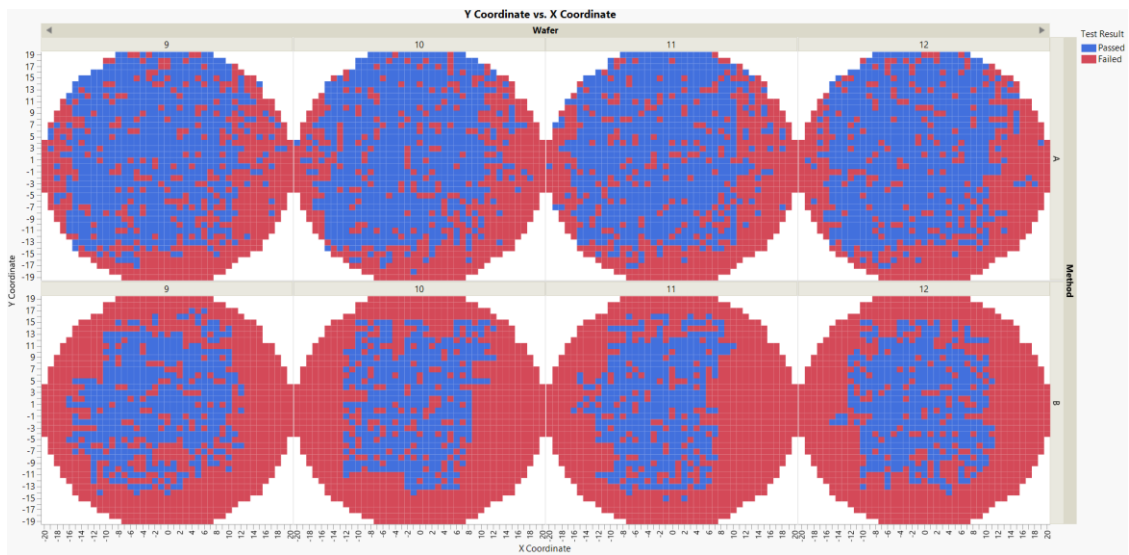
Heatmap of all wafers of Method A and Method B



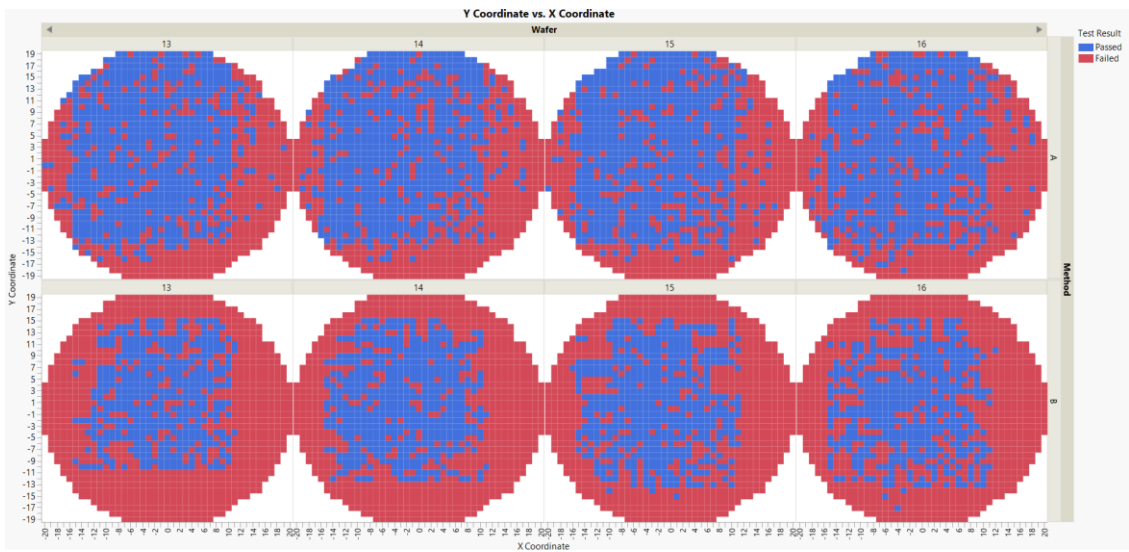
Heatmap of wafers 1 to 4 of Method A and Method B



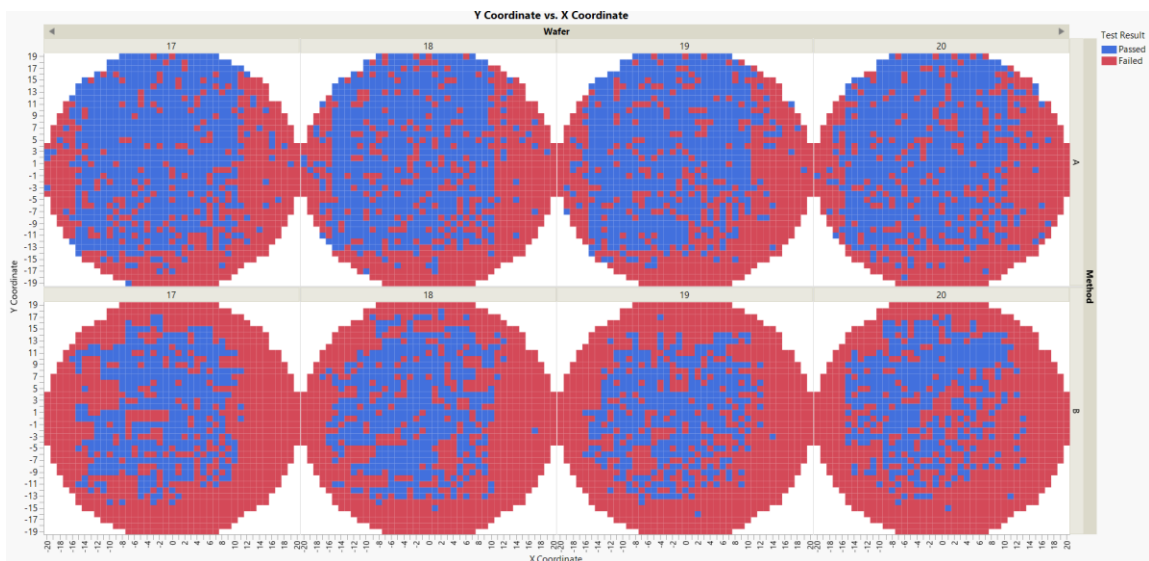
Wafers 5 to 8 Heatmap (Method A and B)



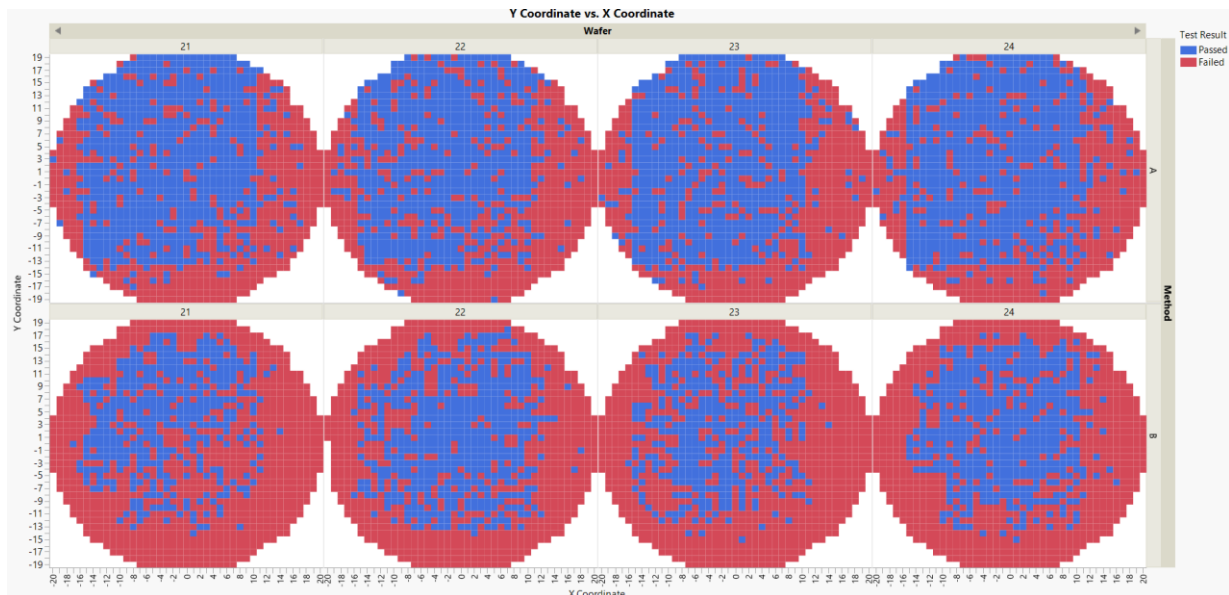
Wafers 9 to 12 Heatmap (Method A and B)



Wafers 13 to 16 Heatmap for method A and B



Wafers 17 to 20 Heatmap for Method A and B



Wafers 21 to 24 Heatmap for Method A and B

Across almost every wafer, **Method A has a larger blue/pass region than Method B**. Method A usually has a broad central passing region, while Method B has a smaller central passing region surrounded by a larger red failed region.

So the heat map confirms the numerical result:

- Method A yield = **56.57%**
- Method B yield = **34.86%**

The key spatial pattern is:

Yield is highest near the center of the wafer and lowest near the outer edge/periphery. This edge-related failure pattern is present in both methods, but it is much more severe for Method B.

The heat maps show that the lower yield of Method B is not caused by one or two bad wafers. Instead, Method B shows a **systematic spatial failure pattern** across nearly all wafers.

The pattern is:

Region	Yield behavior
Center region	Highest yield
Middle region	Mixed pass/fail
Edge/periphery	Lowest yield

Method A also has edge-related failures, but Method B has a much larger failed edge region and a smaller passing center region

The wafer-level heat maps show a consistent spatial yield pattern across the wafer groups. For Method A, most wafers have a broad central passing region, while failures are mainly concentrated near the wafer perimeter, especially along the lower and side edges. In contrast, Method B shows a much larger failed region across nearly all wafers, with passing ICs mostly confined to a smaller central region. This pattern is repeated from wafers 1–25, indicating that the lower yield of Method B is systematic rather than caused by only a few outlier wafers. Overall, the

heat maps suggest that yield is highest near the wafer center and lowest near the wafer edge, with Method B showing much stronger spatial nonuniformity and edge-related yield loss than Method A.

Center vs Edge Yield Analysis

We will add two new columns in JMP:

- 1. Radial Distance
- 2. Wafer Region

Radial distance tells how far each IC is from the wafer centre: $\text{Radial Distance} = \sqrt{X^2 + Y^2}$

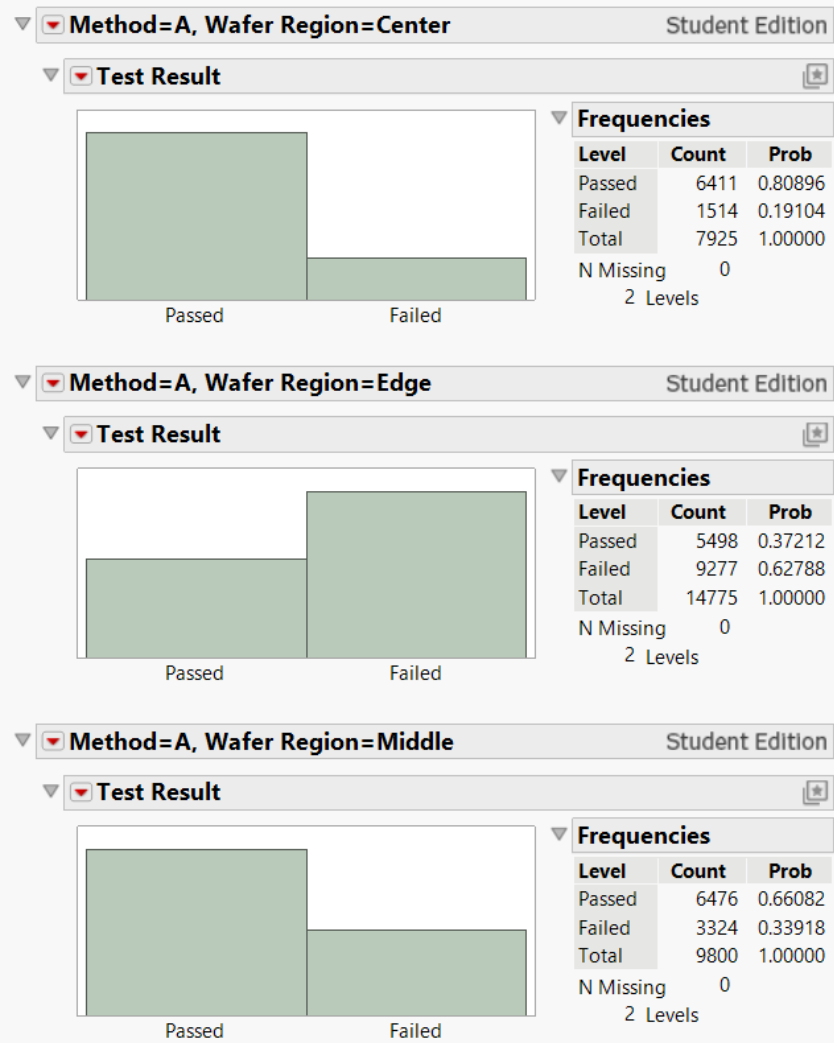
Formula for Column Radial Distance

Region	Meaning
Center	near wafer center (Radial Distance <= 10)
Middle	middle wafer area (Radial Distance <=15)
Edge	near wafer perimeter

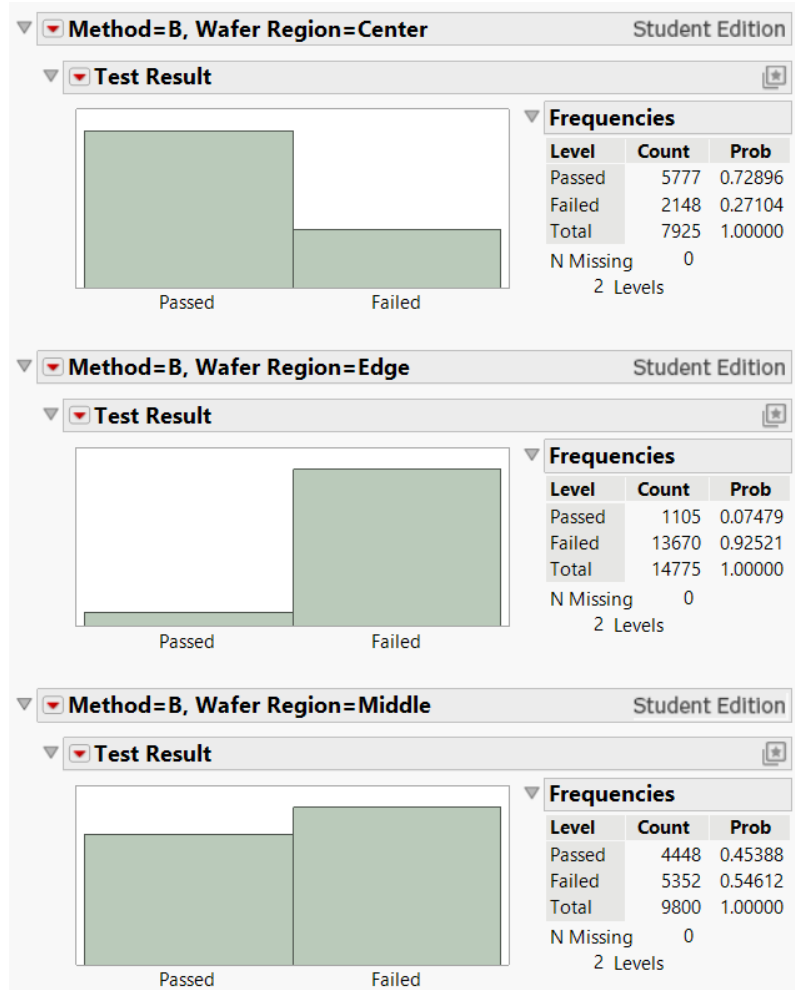
Wafer region formula

Distribution Analysis: Y column: Test Result

By column: Method, Wafer region



Method A Wafer Region Distribution Analysis



Method B Wafer Region Distribution Analysis

Wafer Region	Passed	Failed	Yield
Center	6,411	1,514	80.90%
Middle	6,476	3,324	66.08%
Edge	5,498	9,277	37.21%

Method A Summary

Wafer Region	Passed	Failed	Yield
Center	5,777	2,148	72.90%
Middle	4,448	5,352	45.39%
Edge	1,105	13,670	7.48%

Method B Summary

For both methods: Center Yield > Middle Yield > Edge Yield

Method A performs well in the center region, with a yield of **80.90%**. The yield drops to **66.08%** in the middle region and then drops sharply to **37.21%** at the edge.

This means Method A still has an edge-related yield problem, but it keeps reasonably good yield in the center and middle regions. The center-to-edge yield drop for Method A is: $80.90\% - 37.21\% = 43.69\%$
So Method A loses about **43.69 percentage points** from center to edge.

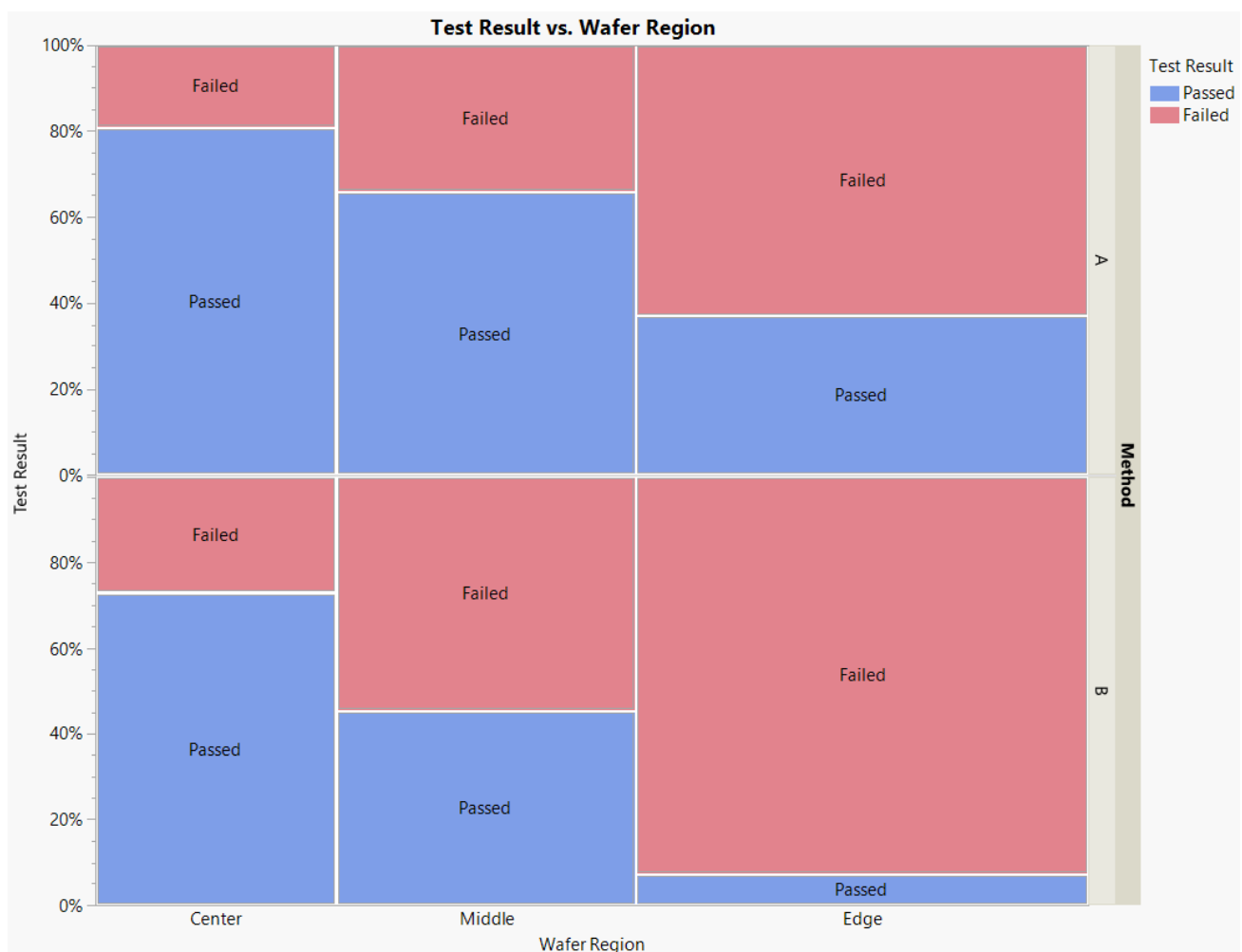
Method B also has its best yield in the center, but even there it is lower than Method A.

Method B center yield is **72.90%**, middle yield is **45.39%**, and edge yield is only **7.48%**.

That edge yield is extremely low. The center-to-edge yield drop for Method B is: $72.90\% - 7.48\% = 65.42\%$

So Method B loses about **65.42 percentage points** from center to edge. This confirms that Method B has a much more severe edge-yield problem.

Region	Method A Yield	Method B Yield	Method A Advantage
Center	80.90%	72.90%	+8.00 percentage points
Middle	66.08%	45.39%	+20.69 percentage points
Edge	37.21%	7.48%	+29.73 percentage points



Mosaic Plot of Wafer region Yield

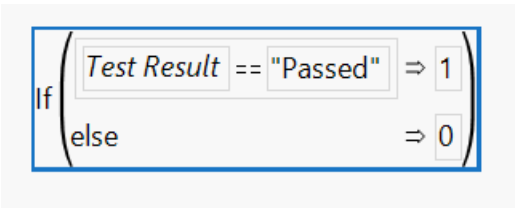
To quantify the spatial yield pattern observed in the wafer heat maps, each IC was classified into Center, Middle, or Edge regions based on radial distance from the wafer center. Then a regional yield analysis was performed. For

Method A, yield decreased from 80.90% in the center to 66.08% in the middle region and 37.21% at the edge. For Method B, the decrease was more severe, with yield dropping from 72.90% in the center to 45.39% in the middle region and only 7.48% at the edge. This confirms a strong center-to-edge yield gradient in both methods. Method A outperformed Method B in every wafer region, with the largest difference occurring at the edge, where Method A exceeded Method B by 29.73 percentage points. These results indicate that Method B suffers from much stronger edge-related yield loss and spatial nonuniformity. The mosaic plot shows a clear center-to-edge yield gradient for both manufacturing methods.

Best and Worst Wafer Ranking

identify the **strongest and weakest wafers** based on wafer-level yield.

New Column : Pass Indicator



Create Wafer-Level yield summary table

Role	Column
Group	Method
Group	Wafer
Statistics	Mean of Pass Indicator
Statistics	N Rows

		Method	Wafer	N Rows	Mean(Pass Indicator)	
1	A	1	1300	0.5730769231		
2	A	2	1300	0.5653846154		
3	A	3	1300	0.5823076923		
4	A	4	1300	0.5938461538		
5	A	5	1300	0.5730769231		
6	A	6	1300	0.5715384615		
7	A	7	1300	0.5869230769		
8	A	8	1300	0.5715384615		
9	A	9	1300	0.5823076923		
10	A	10	1300	0.5807692308		
11	A	11	1300	0.5823076923		
12	A	12	1300	0.5830769231		
13	A	13	1300	0.5646153846		
14	A	14	1300	0.5530769231		

Table Summary of Wafer-Level Yield

Sorted the table summary by wafer yield, in descending and ascending order and then found out the Best wafer and worst wafer overall and also for each method A and B

Category	Method	Wafer	Wafer Yield	Approx. Passed ICs
Best overall wafer	A	4	59.38%	772 / 1300
Worst overall wafer	B	4	25.69%	334 / 1300
Best Method A wafer	A	4	59.38%	772 / 1300
Worst Method A wafer	A	22	53.62%	697 / 1300
Best Method B wafer	B	8 / 25	41.00%	533 / 1300
Worst Method B wafer	B	4	25.69%	334 / 1300

Wafer Level Yield Summary

A wafer-level ranking analysis was performed by converting the pass/fail result into a numeric pass indicator, where Passed = 1 and Failed = 0. The average pass indicator for each wafer was then used as the wafer-level yield. The best overall wafer was Method A Wafer 4, with a yield of 59.38%, while the worst overall wafer was Method B Wafer 4, with a yield of 25.69%. Within Method A, yields ranged from 53.62% to 59.38%, showing relatively stable wafer-to-wafer performance. Within Method B, yields ranged from 25.69% to 41.00%, indicating lower and more variable wafer performance. Notably, the worst Method A wafer still had a higher yield than the best Method B wafer, confirming that Method B's lower yield is a systematic process-level issue rather than the result of only a few outlier wafers.

Engineering Recommendation

Based on the analysis, Method A should be preferred over Method B for further production development. Method A achieved a higher overall yield of 56.57%, compared with 34.86% for Method B. The statistical comparison showed that Method A's yield was about 21.71 percentage points higher than Method B, and the confidence interval confirmed that this difference is significant. Wafer-level analysis showed that Method A was more stable across wafers, while Method B had consistently lower wafer yields. The spatial wafer maps and regional yield analysis further showed that both methods suffer from edge-related yield loss, but the issue is much more severe for Method B. Therefore, Method B should not be moved to full-scale manufacturing without further optimization, especially to improve edge-region yield.